PREP-RASHAN

A Unit of Prep-Rashan Tutorial

DAILY CURRENT AFFAIRS - RASHAN

Comprehensive Compilation for UPSC Civil Services Examination

3 November 2026

General Studies Paper I • Paper II • Paper III • Paper IV

Keywords Defined • PYQ (Prelims & Mains) • Practice Questions (Prelims & Mains)

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GENERAL STUDIES PAPER I

[GS-I] Indian Heritage & Culture, History, Geography of the World and Society

GS-I | Geography – Climatology (Monsoon)

1. Why a Strong El Niño Does Not Always Mean Drought in India

Syllabus: *GS-I: Important geophysical phenomena; climatology – the Indian monsoon. GS-III: Agriculture and disaster management.*

CONTEXT

A **powerful El Niño** is building in the Pacific – **sea-surface warming close to 2°C** – and the **2026 south-west monsoon** ended with a **deficit of about 13%**. Yet past records show that even a **strong El Niño does not guarantee a failed monsoon** in India.

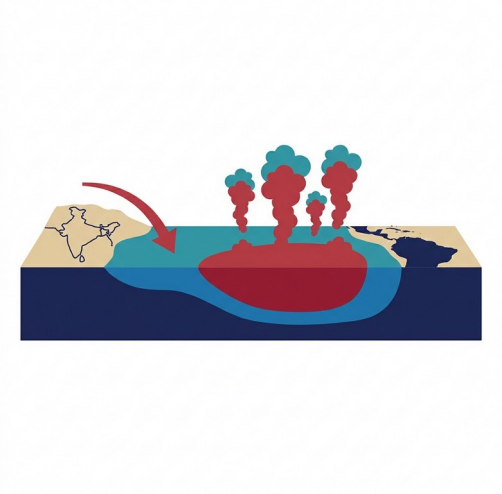


Illustration: warm water and storm clouds over the central Pacific, while dry air sinks over India – the El Niño link, with the Indian Ocean as a possible cushion.

KEYWORDS DEFINED

Keyword	Meaning
El Niño	Periodic **warming of the central and eastern equatorial Pacific**, when the **trade winds weaken**; it often **weakens the Indian monsoon**.
Walker circulation	An **east–west loop of air** over the tropical Pacific: rising air over the warm west, sinking air over the cool east.
Indian Ocean Dipole (IOD)	A **difference in sea-surface temperature** between the **western and eastern Indian Ocean**; a **positive IOD** usually **helps** the monsoon.
Madden–Julian Oscillation (MJO)	An **eastward-moving belt of clouds and rain** that circles the tropics in about **30–60 days**.
Long Period Average (LPA)	The **average all-India monsoon rainfall** over a long base period, used to judge a season as normal, deficient or excess.

How El Niño dries India

- **Normal year:** trade winds push **warm water west** towards Indonesia; air rises there and brings rain, while **cold water upwells** off South America.
- **El Niño year (every 2–7 years):** the winds **weaken or reverse**, the warm pool **moves east**, and the **zone of storms shifts** to the central/eastern Pacific.
- **Effect on India:** to balance the rising air there, air **sinks over the Indian subcontinent** – it **warms and dries** the atmosphere, **suppresses clouds** and **weakens the monsoon winds**.

The record (as cited)

- Of the **seven** moderate-to-strong El Niño years after **1950**, **five** saw a **deficient** monsoon.
- **1997:** a strong El Niño (**+1.64°C**), yet India got **normal rain**; **2015** and **2026** saw **big deficits**.
- **2026:** rainfall about **87.4% of the LPA**, with the **East and North-East** worst hit.
- About **40% of the net sown area** is **rain-fed**, so farm output and rural incomes depend on the monsoon.

Why El Niño is not destiny

- **Positive IOD:** a warmer **western Indian Ocean** sends **extra moisture** towards India and can **offset El Niño** – as in **1997**. In **2015** the dipole **did not turn positive**.
- **Where the Pacific warms:** warming in the **far east (near South America)** keeps the sinking air **away from South Asia**; warming in the **central Pacific (El Niño Modoki)** places it **over India** – a higher drought risk.
- **MJO pulses:** in **August 2023** the MJO **stayed away** from the Indian Ocean and India had its **driest August in over a century**; a late **IOD shift** revived September rain.
- **Local systems: highs over the Iranian plateau** or **lows over the Arabian Sea** can pull moisture inland and **override** the Pacific signal (e.g., **1983, 1997**).

Impacts and way ahead

- **Impacts:** poor **kharif sowing** (paddy, pulses, oilseeds), **low reservoirs and groundwater**, **food inflation** and **export curbs**; forecasting stays hard because ENSO, IOD and MJO interact.
- **Way ahead:** forecasting models that weigh the **Indian Ocean and MJO**, **drought-tolerant and short-duration** crops such as **millets and pulses**, **farm ponds and recharge structures**, and faster **PM Fasal Bima Yojana** payouts using **satellite and drone** data.
- **Value addition:** IMD calls a season **deficient** when rain is **below 90% of the LPA** and **normal** at **96–104%**. **La Niña** – the cool phase – usually **helps** the Indian monsoon.

PYQ – PRELIMS

- **Pattern:** Prelims asks how **El Niño, La Niña and the Indian Ocean Dipole** affect the **Indian monsoon**, and what the **Walker circulation** is.

PYQ – MAINS

- **Pattern:** Mains asks about the **mechanism of the monsoon**, **teleconnections** such as ENSO, and how **climate variability** affects agriculture.

FQ – PRELIMS (Possible Question)

- Consider: 1. A positive Indian Ocean Dipole usually helps the Indian monsoon. 2. In an El Niño year, air rises strongly over the Indian subcontinent. 3. El Niño Modoki refers to peak warming in the central Pacific. Which are correct? **(Ans: 1 and 3 only)**

FQ – MAINS (Possible Question)

- "El Niño is a risk factor for the Indian monsoon, not a guarantee of drought." Explain with reference to the Indian Ocean Dipole and intraseasonal oscillations. (150 words)

GS-I | Geography – Cryosphere & Climate Change

2. State of the Cryosphere 2026: The Price of Overshooting 1.5°C

Syllabus: *GS-I: Important geophysical phenomena; changes in critical geographical features (glaciers, ice caps). GS-III: Environment – climate change; disaster management (GLOFs).*

CONTEXT

Ahead of **COP31 in Antalya, Türkiye**, the **International Cryosphere Climate Initiative (ICCI)** and **UNEP** released two reports warning that even a **temporary overshoot of 1.5°C** could cause **irreversible ice loss** and a large **release of carbon from permafrost**.



Photo: the snout of the Gangotri Glacier (Gaumukh), source of the Bhagirathi – one of the Himalayan glaciers in retreat.
Photo: gbSk at Flickr, Wikimedia Commons, CC BY 2.0

KEYWORDS DEFINED

Keyword	Meaning
Cryosphere	All parts of Earth where water is **frozen** – **ice sheets, glaciers, snow, sea ice, lake and river ice and permafrost**.
Overshoot	A pathway where warming **temporarily goes above a limit** (here **1.5°C**) before being brought back down.
Permafrost	Ground that stays **frozen for at least two years in a row**; it holds huge amounts of **carbon**.
Albedo	The **share of sunlight a surface reflects** – fresh snow and ice reflect most of it.
GLOF	**Glacial Lake Outburst Flood** – a sudden flood when a **lake dammed by ice or moraine** bursts.

Key findings (as cited)

- The **Arctic** is warming about **four times faster** than the world; the **Hindu Kush Himalaya** warms by about **1.8–2.2°C for every 1.5°C** of global warming.
- **Permafrost** covers about **14 million sq km** and stores about **1,700 billion tonnes of carbon**.
- **Greenland and Antarctica** are losing ice about **six times faster** than in the **1990s**.

- With **2.8–3.2°C** of warming, sea level could rise about **5 cm a year by 2150** – faster than coastal wetlands can adjust.
- At **3°C**, most mid-latitude glaciers **below 6,000 m** would vanish and about **two-thirds of Hindu Kush Himalaya ice** would be lost.
- Thawing permafrost could emit, each year for **100–200 years**, about as much as the **USA emits today**.
- **India:** about **6,119 deaths in 62 GLOFs between 1835 and 2025**, often set off by **ice avalanches and monsoon rain**.

Why the ice is shrinking

- **Greenhouse gases above 420 ppm** keep warming the poles and mountains; Greenland's **tipping point** may lie close to **1.5°C**.
- **Albedo feedback:** less snow and sea ice exposes **dark sea and rock**, which absorb more heat.
- **Warm ocean water** melts **ice shelves from below**, weakening West Antarctica's glaciers.
- **Rain on snow, heavy downpours and wildfires** speed up melt and permafrost thaw.

What it means

- **Higher seas** threaten **coastal cities and deltas**.
- **Less meltwater** in the dry season for the **Indus, Ganga and Brahmaputra** – risks for **farms, drinking water and hydropower**.
- **More GLOFs and landslides** in the Himalaya.
- **Permafrost turning into a carbon source** and a **weaker AMOC**, which can disturb **monsoons and European weather**.

What the reports ask for

- **Cap peak warming near 1.7°C** for no more than about **two decades**, reach **net-zero by 2050**, and bring warming back down afterwards.
- Pair emission cuts with **carbon dioxide removal** to bring CO_2 back towards **350–380 ppm**.
- **Early warning for glacial lakes** – water-level sensors, **siphoning of lakes** and **satellite (InSAR) monitoring**.
- **Adaptation and loss-and-damage finance** for **mountain and Arctic communities**.
- **Value addition:** the **South Lhonak lake GLOF (Sikkim, October 2023)** destroyed the **Teesta-III dam**; the **NDMA** has guidelines on managing GLOFs.

PYQ – PRELIMS

- **Pattern:** Prelims asks about **glaciers, permafrost, albedo, sea ice** and phenomena such as **GLOFs**.

PYQ – MAINS

- **Pattern:** Mains asks about **glacier retreat in the Himalaya**, its effect on **water security**, and **disaster preparedness** for GLOFs.

FQ – PRELIMS (Possible Question)

- Consider: 1. Permafrost is ground that stays frozen for at least two consecutive years. 2. Loss of sea ice raises Earth's albedo. 3. A GLOF is a flood caused by the bursting of a glacial lake. Which are correct? **(Ans: 1 and 3 only)**

FQ – MAINS (Possible Question)

- "For the Himalayan region, overshooting 1.5°C is not a climate statistic but a survival risk." Discuss with reference to the State of the Cryosphere 2026 report. (250 words)

GS-I | Geography – Protected Areas of Nepal (Mapping)

3. Shey-Phoksundo: Nepal's Largest National Park and Its Turquoise Lake

Syllabus: *GS-I: Location of places – important geographical features of the neighbourhood (mapping). GS-III: Conservation – protected areas and wetlands.*

CONTEXT

Six Indian trekkers caught by **heavy snowfall** on high passes (**Numla and Wagla**) near **Shey-Phoksundo Lake** in Nepal were **airlifted to safety by a private helicopter**; they had been **unreachable for two days**.



Photo: Shey-Phoksundo Lake in Lower Dolpo, Nepal – famous for its turquoise water.
Photo: Outlaw Django, Wikimedia Commons, CC BY-SA 4.0

KEYWORDS DEFINED

Keyword	Meaning
Trans-Himalaya	The **dry, high region north of the main Himalayan range**, in the rain shadow, close to the Tibetan Plateau.
Oligotrophic lake	A lake with **very few nutrients**, so water is **clear** and plant growth is low.
Landslide-dammed lake	A lake formed when a **landslide blocks a river valley**.
Ramsar site	A **wetland of international importance** listed under the **Ramsar Convention (1971)**.
Buffer zone	An area **around a protected area** where local people use resources under **co-management**.

The national park

- **Nepal's largest national park** (about **3,555 sq km**) and its **only protected area in the trans-Himalayan zone**.
- In **Dolpa and Mugu districts** of **Karnali Province**, north-west Nepal, about **300 km** from Kathmandu; headquarters at **Palam (Dolpa)**.
- **Set up in 1984**; a **216 sq km buffer zone** was added in **2006** for **community co-management**.
- **Height range:** from about **2,130 m** to **6,883 m** at **Kanjiroba Himal**.

Phoksundo Lake

- An **alpine freshwater lake** at about **3,612 m**, formed by a **landslide dam** some **30,000–40,000 years ago**.
- Area about **494 hectares**, volume about **409 million cubic metres**, deepest point about **136 m** (2019 survey).
- Water spills over a **167-metre waterfall** below the natural dam.
- **Oligotrophic** (low in nutrients); a **Ramsar site since 2007**; **20+ Tibetan stupas** line its northern side.

Wildlife and culture

- Home to the **snow leopard**, with prey such as **blue sheep, Himalayan musk deer and argali**.
- **Dry rain-shadow** slopes with **juniper and shrubs**; moist southern valleys with **birch (Bhojpatra) and silver fir**.
- **Shey Gompa** and **Ringmo village** keep alive **Tibetan Buddhist** and the older **Bon** traditions.
- **Value addition:** the **Dolpo** region is the setting of Peter Matthiessen's travel classic **'The Snow Leopard'**.

Lessons from the rescue

- High passes in **late autumn** can close suddenly with **snow**; trekkers need **local guides, permits and weather checks**.
- Highlights **India–Nepal cooperation** in **search and rescue** for the many Indians who trek in Nepal.

PYQ – PRELIMS

- **Pattern:** Prelims asks about **lakes, national parks and Ramsar sites** of India and its **neighbours** (mapping).

PYQ – MAINS

- **Pattern:** Mains (GS-I) asks about **Himalayan landforms**, and GS-III about **ecotourism and protected-area management**.

FQ – PRELIMS (Possible Question)

- Shey-Phoksundo Lake, recently in the news, is located in: (a) Sikkim (b) Bhutan (c) Nepal (d) Ladakh **(Ans: c)**

FQ – MAINS (Possible Question)

- "High-altitude tourism needs to balance livelihoods, safety and fragile ecosystems." Discuss with examples from the Himalaya. (150 words)

GS-I | Indian Culture – Traditional Knowledge Systems

4. CDTK-2026: Sharing India's Science-Tested Traditional Knowledge

Syllabus: *GS-I: Indian culture – salient aspects; Indian knowledge systems. GS-III: Science and technology – awareness; IPR issues.*

CONTEXT

CDTK-2026 – the **third edition** of an international conference on **communicating and spreading traditional knowledge** – opened on **2 November 2026** at **CSIR-NPL, New Delhi**, under the **SVASTIK** initiative.



Illustration: a palm-leaf manuscript beside a laptop and a microscope, with herbs, millets, metalware and temple architecture – old knowledge tested by modern science.

KEYWORDS DEFINED

Keyword	Meaning
Traditional knowledge (TK)	Knowledge, skills and practices **developed by communities over generations** – in medicine, farming, food, crafts and more.
SVASTIK	**Scientifically Validated Societal Traditional Knowledge** – a CSIR-NIScPR initiative (2021).
CSIR-NIScPR	**CSIR-National Institute of Science Communication and Policy Research**, New Delhi.
Scientific temper	Thinking based on **evidence, reason and enquiry** – a **Fundamental Duty** under **Article 51A(h)**.
Biopiracy	**Patenting or commercial use of traditional knowledge or bio-resources** without consent or benefit-sharing.

About CDTK-2026

- An **international conference** held under **SVASTIK** to promote **traditional knowledge** (described as **biennial**).
- **Organiser: CSIR-NIScPR**; **venue: CSIR-National Physical Laboratory (NPL), New Delhi**.
- **Earlier editions: CDTK-2023** in New Delhi; **CDTK-2024** with **Gurugram University**.

- **Aim:** bring together **experts, researchers and knowledge holders** to share how TK can be **communicated and spread**.

What it covers

- Traditional knowledge in **medicine, agriculture, cuisine, metallurgy, architecture, mathematics, ecology, environment and education**.
- **Participants:** scholars, practitioners, **cultural ambassadors**, **knowledge holders** and **grassroots innovators**.
- **Method:** science-tested practices are shared through **digital and print media** in **Indian and foreign languages**.

About SVASTIK

- **Launched in August 2021** by **CSIR-NIScPR**.
- **Aims: conserve traditional knowledge**, promote **scientific temper**, and build **public confidence** in its scientific value.
- Only knowledge that has been **scientifically validated** is promoted – separating **evidence from myth**.

Why it matters

- Protects TK from **biopiracy** and links it to **health, nutrition and sustainability**.
- **Value addition:** India's **Traditional Knowledge Digital Library (TKDL, 2001)** – by **CSIR** and the **Ministry of AYUSH** – helps patent offices reject claims on known knowledge, as in the **turmeric (1997)** and **neem** patent cases. **CSIR-NIScPR** was formed in **2021** by merging **NISCAIR and NISTADS**.

PYQ – PRELIMS

- **Pattern:** Prelims asks about **TKDL, GI tags, biopiracy** and **CSIR institutions**.

PYQ – MAINS

- **Pattern:** Mains asks how India can **protect and use traditional knowledge** while avoiding **unscientific claims**.

FQ – PRELIMS (Possible Question)

- SVASTIK, recently in the news, stands for: (a) Scheme for Village Agriculture and Traditional Knowledge (b) Scientifically Validated Societal Traditional Knowledge (c) Society for Vedic Arts and Science (d) Sustainable Village Technology Kit **(Ans: b)**

FQ – MAINS (Possible Question)

- "Traditional knowledge becomes a national asset only when it is scientifically validated and legally protected." Discuss. (150 words)

GENERAL STUDIES PAPER II

[GS-II] Governance, Constitution, Polity, Social Justice & International Relations

GS-II | International Relations – UN Peacekeeping **UPDATE**

5. MINURSO: Why the Western Sahara Mission Ended – Vetoes and Vacuum

Syllabus: *GS-II: Important international institutions – UN Security Council and peacekeeping; effect of policies of other countries on India's interests.*

WHAT WE KNOW TILL NOW

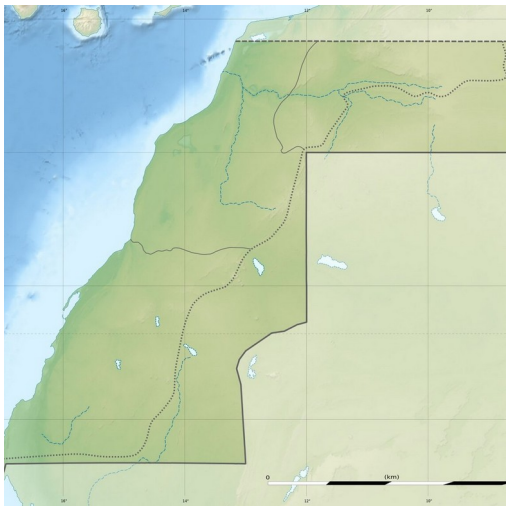
- **MINURSO**, the **UN Mission for the Referendum in Western Sahara**, was set up in **1991** after the **Morocco–Polisario ceasefire**; its headquarters is at **Laayoune**.
- **Morocco controls about 80%** of the territory; the **Polisario Front** seeks **self-determination**; the **referendum was never held**.
- The mandate **ended at the close of October** when the **Security Council did not renew it**, and the **wind-up (liquidation)** began.

WHAT TODAY ADDS

- **Why it lapsed: Russia and the United States vetoed each other's draft resolutions**, so no renewal could pass.
- Details of the mission's **work, history and the renewal process** – nine team sites, mine action, the Tindouf office and the **9-votes-no-veto** rule.

CONTEXT

The **mandate of MINURSO expired on 31 October 2026** after **Russia and the United States vetoed each other's competing draft resolutions** in the **UN Security Council**, leaving no mission to watch the **Western Sahara** conflict.



Map: Western Sahara on the Atlantic coast of north-west Africa, between Morocco, Algeria and Mauritania.

Photo: Eric Gaba (Sting - fr:Sting) and user:NordNordWest, Wikimedia Commons, CC BY 3.0

KEYWORDS DEFINED

Keyword	Meaning
Penholder	The **Security Council member that drafts** resolutions on a given issue – the **USA** for Western Sahara.
Veto	The power of any of the **five permanent members (P5)** to **block** a substantive Council resolution.
The berm	A **long sand wall** built by **Morocco** across Western Sahara, separating the areas it controls from the Polisario side.
UNMAS	The **UN Mine Action Service**, which clears **landmines and unexploded ordnance**.
Group of Friends	A small group of countries that **advises the penholder** on a conflict at the UN.

History

- **Set up on 29 April 1991** by **UNSC Resolution 690** to carry out the **settlement plan** accepted by **Morocco and the Polisario Front**.
- **Referendum stalled:** the two sides could not agree **who is eligible to vote**, so MINURSO mainly **watched the ceasefire**.
- **2020:** after the **Guerguerat crisis**, the Polisario said the ceasefire was **over** and resumed **limited fighting**; MINURSO kept monitoring.

What MINURSO did

- **Military observers** at **nine team sites** on **both sides of the berm** – patrols, checking violations, lowering tension.
- **Mine action** with **UNMAS** – clearing **mines and unexploded ordnance** for local and nomadic people.
- **Support to the UN envoy** in talks involving **Morocco, the Polisario, Algeria and Mauritania**.
- **Liaison:** headquarters at **Laayoune**, a liaison office at **Tindouf (Algeria)**, working with agencies helping **Sahrawi refugees**.
- **Confidence-building:** family visits and contact between **refugees and communities** across the territory.

How renewal works

- The **USA, as penholder**, drafts the resolution after consulting the **Group of Friends of Western Sahara** and other members.
- The **UN Secretary-General** reports on security, ceasefire breaches, humanitarian needs and politics.
- A renewal needs **at least 9 of 15 votes** and **no veto** from the **P5 (USA, UK, France, Russia, China)**.

Why it matters

- Shows how **P5 rivalry** can end a mission **without peace on the ground** – a test of **UN credibility**.
- **Value addition:** most **Sahrawi refugees** live in camps near **Tindouf**; the **Sahrawi Arab Democratic Republic** is a member of the **African Union**. India backs a **UN-led political solution**.

PYQ – PRELIMS

- **Pattern:** Prelims asks about the **UN Security Council**, the **veto**, and **disputed territories** in Africa and West Asia.

PYQ – MAINS

- **Pattern:** Mains asks about **UNSC reform** and whether **great-power rivalry** paralyses the Council.

FQ – PRELIMS (Possible Question)

- With reference to MINURSO, consider: 1. It was set up by UNSC Resolution 690 in 1991. 2. Its renewal needs a simple majority of the UN General Assembly. 3. It had a liaison office at Tindouf in Algeria. Which are correct? **(Ans: 1 and 3 only)**

FQ – MAINS (Possible Question)

- "The lapse of MINURSO shows the cost of veto politics in the Security Council." Examine, and suggest reforms. (150 words)

GS-II | Social Justice – Child Protection Online

6. ScrollControl: IAF's Child Digital Safety Programme in Its Schools

Syllabus: *GS-II: Issues relating to children; education. GS-III: Cyber security. GS-IV: Ethics of technology and protection of the vulnerable.*

CONTEXT

The **Indian Air Force** has rolled out the **'ScrollControl' Child Digital Safety Programme** in **Air Force Schools**; **261 teachers from 42 schools** are now certified as **Cyber Crime Intervention Officers**.



Illustration: a teacher shows pupils a tablet with a shield, blocking icons of online risks – grooming, gaming traps and money fraud.

KEYWORDS DEFINED

Keyword	Meaning
Online grooming	When an adult **builds trust with a child online** in order to **exploit or abuse** them.
Cyber hygiene	Everyday **safe habits online** – strong passwords, privacy settings, not sharing personal details.
Digital citizenship	Using technology **responsibly, safely and respectfully**.
Cyber Crime Intervention Officer	A **trained teacher** who can **spot and respond to** cyber threats faced by students.
Deepfake	**AI-made fake audio, image or video** that looks real – a new tool for bullying and fraud.

What the programme is

- A **cyber-safety and digital-literacy** initiative adopted by the **IAF** for its schools.
- **Aim:** protect children from **online threats** and build **responsible digital behaviour** through **trained teachers**.

Key features

- **Teacher certification: 261 teachers in 42 Air Force Schools** trained as **Cyber Crime Intervention Officers**.
- **Focus areas: online grooming, cyber exploitation and misuse of AI**.

- **Gaming and money traps:** awareness about **risky games** and **online financial fraud**.
- **School-level framework:** cyber awareness and **response capacity** built into each school.

Why it matters

- Children go online **younger and longer**; schools are the natural **first line of defence**.
- **Early detection** – trained teachers can spot **bullying, grooming or fraud** before harm grows.
- Builds **cyber hygiene and digital literacy** for safer use of **interconnected platforms**.

Legal and institutional support

- **POCSO Act, 2012** protects children from sexual offences, including online.
- **IT Act, 2000 (Section 67B)** punishes material showing **children in sexually explicit acts**.
- **DPDP Act, 2023** requires **verifiable parental consent** to process children's data and bars **tracking and targeted ads** aimed at them.
- **Value addition:** complaints can be made on **cybercrime.gov.in** or helpline **1930**, run under the **Indian Cyber Crime Coordination Centre (I4C)** of the **Home Ministry**.

PYQ – PRELIMS

- **Pattern:** Prelims asks about **cyber-security institutions (CERT-In, I4C)** and laws such as **POCSO and the IT Act**.

PYQ – MAINS

- **Pattern:** Mains asks about **children's online safety**, **social media regulation** and **cyber crime**.

FQ – PRELIMS (Possible Question)

- Consider: 1. The DPDP Act, 2023 requires verifiable parental consent for processing children's data. 2. Helpline 1930 is meant for reporting cyber fraud. 3. Section 67B of the IT Act deals with online material involving children. Which are correct? **(Ans: All three)**

FQ – MAINS (Possible Question)

- "Schools must become the first line of defence for children in the digital world." Discuss with reference to recent initiatives. (150 words)

GS-II | Governance – Rural Employment & Grievance Redressal UPDATE

7. VB-G RAM G: New Rules Promise Grievance Redressal in 15 Days

Syllabus: *GS-II: Government policies and interventions; welfare schemes for vulnerable sections; transparency and accountability.*

WHAT WE KNOW TILL NOW

- The **VB-G RAM G Act, 2025 replaced MGNREGA** and promises **125 days** of wage work a year to rural households.
- A **new list of 375 permissible works** spans **water security, core rural infrastructure, livelihoods and disaster preparedness**.
- Works are planned through **Viksit Gram Panchayat Plans** approved by the **Gram Sabha**.

WHAT TODAY ADDS

- The **Grievance Redressal Rules, 2026**: complaints must be **settled within 15 days**, with **inquiry in 7 working days**.
- **Every complaint must be registered and digitised** with a **tracking number**; two draft clauses were **dropped**.

CONTEXT

The **Ministry of Rural Development** has notified the **Grievance Redressal Rules, 2026** under **VB-G RAM G** (the **Viksit Bharat – Guarantee for Rozgar and Ajeevika Mission (Gramin)**), requiring **every scheme complaint to be resolved within 15 days**.

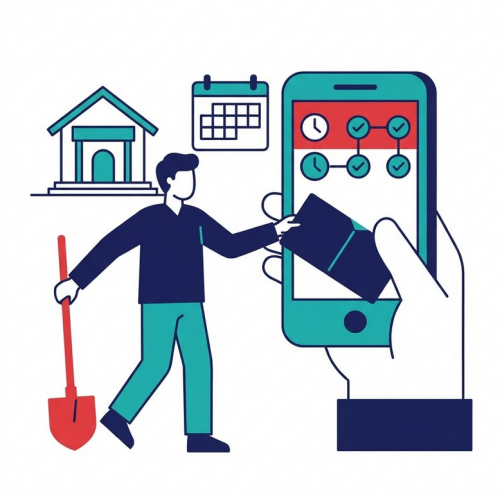


Illustration: a rural worker files a complaint on a phone that shows a tracking timeline – time-bound redressal.

KEYWORDS DEFINED

Keyword	Meaning
Grievance Redressal Officer (GRO)	The official at the **Gram Panchayat, Block or District** level who **receives and decides** complaints.
Muster roll	The **attendance and wage register** of workers on a public-works site.

Keyword	Meaning
Digital Public Infrastructure (DPI)	Shared digital platforms – here a **central grievance portal** – used to **record and track** complaints.
Escalation	Automatic **transfer of an unresolved complaint** to a **higher authority**.
Social audit	**Public verification** of a scheme's spending and works by the **community**, usually through the **Gram Sabha**.

What the rules set up

- A **statutory, multi-tier system** to receive, record, inquire into, track and **dispose of complaints** under the **VB-G RAM G Act, 2025**.
- **Aim:** protect rural households' **right to work and wages** and make local officials **accountable**.

Key features

- **15-day limit:** every complaint – on **work allotment, wage payment, asset quality or muster rolls** – must be **finally resolved within 15 days**.
- **7-day inquiry: GROs** at Gram Panchayat, Block or District level must check records, hear the complainant if needed and **finish the inquiry within 7 working days**.
- **Many ways to complain: in person**, by **Speed Post**, through a **phone helpline** or on the **central portal**.
- **No rejection on technicalities:** a complaint cannot be refused because of its **format, language or mode** of submission.
- **Same-day digitisation:** paper complaints are entered on the **central DPI platform** the **same or next working day**, with an **acknowledgement and tracking number**.
- **Auto-escalation:** complaints pending **beyond 15 days** move up automatically, with an **appeal** under the Act.

What was dropped from the draft

- **Automatic FIRs** for **financial misappropriation**.
- Placing **action-taken reports** before **Gram and District Panchayat meetings**.
- **Debate:** critics say this weakens **deterrence and local scrutiny**; the government stresses **speed and digital tracking**.

Why it matters

- Delayed wages and denied work are the **most common complaints** in rural job schemes – **time limits** build trust.
- **Value addition:** under MGNREGA, **social audits by the Gram Sabha** and district **Ombudspersons** were key accountability tools; strong redressal must go **hand in hand with social audit**.

PYQ – PRELIMS

- **Pattern:** Prelims asks about **features of rural employment schemes** – days of work, funding, social audit, wage payment.

PYQ – MAINS

- **Pattern:** Mains asks about **accountability, social audit and grievance redressal** in welfare schemes.

FQ – PRELIMS (Possible Question)

- Under the VB-G RAM G Grievance Redressal Rules, 2026, consider: 1. Complaints must be resolved within 15 days. 2. A complaint may be rejected if it is not in the prescribed format. 3. Unresolved complaints are escalated automatically. Which are correct? **(Ans: 1 and 3 only)**

FQ – MAINS (Possible Question)

- "Time-bound grievance redressal is the backbone of a rights-based welfare scheme." Examine with reference to the VB-G RAM G rules. (150 words)

GS-II | International Relations – India–Bhutan; Buddhist Heritage

8. Project 108: Bhutan Raises 108 Stupas at Gelephu Mindfulness City

Syllabus: *GS-II: India and its neighbourhood – relations. GS-I: Indian culture – Buddhist art and architecture.*

CONTEXT

Bhutan marked **Lhabab Duchen** by consecrating **108 stupas** in a single day at **Gelephu Mindfulness City (GMC)**; **Stupa No. 76** was dedicated to **India's Prime Minister** in recognition of India's support and **India–Bhutan ties**.



Photo: Chendebji Chorten in central Bhutan – a stupa (chorten) of the kind raised under Project 108.
Photo: Arian Zwegers from Brussels, Belgium, Wikimedia Commons, CC BY 2.0

KEYWORDS DEFINED

Keyword	Meaning
Chorten (stupa)	A **Buddhist monument** that holds **relics or sacred objects**; 'Jangchub Chorten' means **Stupa of Enlightenment**.
Lhabab Duchen	A Bhutanese festival marking the **Buddha's descent from heaven** to the human world.
Vajrayana	The **'Diamond Vehicle'** school of Buddhism followed in **Bhutan and Tibet**.
Zung	**Sacred contents** – scriptures, mantras, relics and prayers – sealed inside a stupa to **consecrate** it.
Je Khenpo	The **Chief Abbot** of Bhutan's central monastic body.

What Project 108 is

- A **royal spiritual and architectural project**: **108 Jangchub Chortens** raised and consecrated **on one day** along an **11.7-km corridor** at **GMC** in **southern Bhutan**.
- **Guided by King Jigme Khesar Namgyel Wangchuck** as the **spiritual foundation** of Gelephu Mindfulness City.
- Held on **1 November 2026**, on **Lhabab Duchen**, close to the King's **18th coronation anniversary**.
- About **40,000 volunteers, monks and artisans** built the stupas in about **nine hours** (as cited).

Key features

- **Sacred spacing:** stupas **108 metres apart** along the **Mau Chhu river**, near the **border with Assam**.

- **Old meets new:** each **15-metre** chorten follows **classical Buddhist proportions** but uses **prefabricated steel frames** and **local materials**.
- **Consecration:** monastic assemblies led by the **Je Khenpo**, with **traditional horns** and **Avalokiteshvara mantras**.
- **Zung** – scriptures, mantras, relics and prayers – sealed inside each stupa.
- **Aim:** make Gelephu a **global Vajrayana centre** that blends **spiritual well-being** with **sustainable development**.

Stupa No. 76 and India

- Dedicated by Bhutan's King to **India's Prime Minister**; the number marks his **entering his 76th year**.
- A tribute to India's **financial, infrastructure and connectivity support** for GMC and to the **India–Bhutan partnership**.

India–Bhutan context

- **Gelephu Mindfulness City:** a **special administrative region** announced by the King in **December 2023**, planned as a green economic hub next to **Assam**.
- **Value addition:** the **India–Bhutan Friendship Treaty (1949, revised 2007)**; India is Bhutan's **largest development and trade partner**, and a **Kokrajhar (Assam)–Gelephu rail link** is planned.

PYQ – PRELIMS

- **Pattern:** Prelims asks about **Buddhist sites, festivals and schools**, and India's **neighbourhood projects**.

PYQ – MAINS

- **Pattern:** Mains asks about **India–Bhutan relations** and the role of **culture and connectivity** in neighbourhood diplomacy.

FQ – PRELIMS (Possible Question)

- Gelephu Mindfulness City, recently in the news, lies close to which Indian state? (a) Sikkim (b) West Bengal (c) Assam (d) Arunachal Pradesh **(Ans: c)**

FQ – MAINS (Possible Question)

- "Shared Buddhist heritage is a strategic asset in India's neighbourhood policy." Discuss with reference to India–Bhutan relations. (150 words)

GENERAL STUDIES PAPER III

[GS-III] Technology, Economic Development, Bio-diversity, Environment, Security & Disaster Management

GS-III | Science & Technology – AI and Data Protection

9. Training AI on Our Data: The Privacy Question India Must Answer

Syllabus: *GS-III: Science and technology – developments in IT and AI. GS-II: Fundamental rights (privacy); government policies and regulation.*

CONTEXT

Privacy is a fundamental right (**Puttaswamy, 2017**). As **AI models are trained on huge amounts of online data**, questions arise about how **personal information is collected and used** without people knowing – and whether India's **data protection law** covers it.



Illustration: photos, locations and profiles funnelled into an AI 'brain', with a padlock for privacy.

KEYWORDS DEFINED

Keyword	Meaning
Training data	The **huge collection of text, images and other content** used to teach an AI model.
Memorisation	When a model **stores pieces of its training data** and can **repeat them** – including personal details.
Data fiduciary	Under the **DPDP Act**, the entity that **decides why and how personal data is processed**.
Right to object	A right (as in the EU's **GDPR**) to **refuse** a particular use of one's data, such as **AI training**.
Privacy by design	Building **privacy safeguards into a system from the start**, not adding them later.

How AI training can hurt privacy

- **Scraping without consent:** firms gather large amounts of **personal information** from websites and social media, often **without informed consent**.
- **Memorisation and leakage:** models may **store sensitive details** and **reveal them** when asked the right questions.
- **Identifying people:** studies show models can be **queried to pull out personal details** and build **profiles**.
- **Sensitive data:** names, gender, **sexual orientation**, workplace, **images, browsing history** and behaviour patterns.
- **Profiling and bias:** misuse can lead to **discrimination** and loss of **autonomy**.

Gaps in India's law

- The **Digital Personal Data Protection (DPDP) Act, 2023** and the **DPDP Rules (notified November 2025)** require **notice, consent and safeguards**.
- **Grey zone:** it is unclear how the law applies to **publicly available data** later used for **AI training**.
- **Erasure is not unlearning:** deleting data does **not remove what a trained model has already absorbed**.
- **No general right to object** to AI training for Indian users – unlike users in the **European Union**, where one major platform offers an **opt-out**.
- **Value addition:** the DPDP Act does **not apply** to personal data that a person has **made publicly available** themselves – the root of the AI-training debate. Penalties go up to **₹250 crore**, imposed by the **Data Protection Board of India**.

Practices of AI companies

- **Is 'public' the same as 'consent'?** – firms treat **publicly visible posts** as fair material for training.
- **Weak accountability:** data may be used for **purposes beyond the original one**, and users rarely know.
- **Commercial profiling:** insights about people can be **sold or used to target** them.

Safeguards needed

- **Clarify the DPDP framework** for AI training, **inferred data** and **public data**.
- **Transparency:** disclose **data sources, purposes and safeguards**.
- **Opt-out and consent** mechanisms that are **easy to use**.
- **Privacy by design and data minimisation**; test models for **memorisation**; use techniques such as **differential privacy**.
- **Accountability:** monitoring, enforcement and penalties; extra care for **vulnerable groups**; **innovation without trading away rights**.

PYQ – PRELIMS

- **Pattern:** Prelims asks about the **right to privacy (Puttaswamy)**, **data protection law** and basic **AI terms**.

PYQ – MAINS

- **Pattern:** Mains asks about **AI's opportunities and risks**, **data protection** and **regulating big tech**.

FQ – PRELIMS (Possible Question)

- Consider: 1. The Supreme Court declared privacy a fundamental right in the Puttaswamy case (2017). 2. The DPDP Act, 2023 gives Indian users a general right to object to the use of their data for AI training. 3. Deleting data from a database automatically removes it from a trained AI model. Which are correct? **(Ans: 1 only)**

FQ – MAINS (Possible Question)

- "AI training models pose a new challenge to the right to privacy." Examine the gaps in India's data protection framework and suggest safeguards. (250 words)

GS-III | Indian Economy – Energy Security & Macroeconomic Stability

10. Oil Shocks: How India Is Learning to Absorb Them – and Grow

Syllabus: *GS-III: Indian economy – growth, inflation, fiscal and monetary policy; energy security; external sector.*

CONTEXT

India **imports most of its crude oil** and was once hit hard by every **Middle East crisis**. Recent shocks show **greater resilience** – thanks to changes in the **energy mix** and **careful fiscal and monetary policy** – and the argument now is to turn shocks into a **chance for reform**.



Illustration: an oil barrel balanced against solar, wind and fiscal savings – resilience through diversification.

KEYWORDS DEFINED

Keyword	Meaning
Energy intensity	**Energy used per unit of GDP** – lower intensity means oil prices hurt growth less.
Inflation targeting	RBI's framework of keeping **CPI inflation at 4% (±2%)**, which anchors expectations.
Second-round effects	When a one-time price jump spreads into **wages and other prices** and becomes **persistent inflation**.
Current Account Deficit (CAD)	When a country **spends more abroad than it earns** on trade, services and income – financed by capital inflows.
Fiscal space	Room in the **budget** to spend or cut taxes **without risking debt sustainability**.

Why India copes better now

- **Lower energy intensity** and **more alternative energy** reduce oil's weight in costs.
- **More supplier regions** and **large refining capacity** cushion supply disruptions.
- **Counter-cyclical excise and pricing** soften the pass-through to consumers.
- **Inflation targeting** limits how far firms **pass on higher costs**.
- **Fiscal space** from post-pandemic consolidation and **better tax revenue**.

- **Real prices:** even if crude crosses **$100 a barrel**, its **inflation-adjusted** level is well **below earlier peaks** such as **2012**.

Smart policy design

- **Supply side first:** keep consolidating while improving the **quality of public spending**.
- **Use buffers wisely: forex reserves**, fiscal space and regulation to absorb **temporary** shocks.
- **Fiscal–monetary coordination:** fiscal policy raises **potential output**; monetary policy keeps actual output close to it and **prevents second-round inflation** – without hurting **RBI's independence**.
- **No excess liquidity**; **pass through persistent shocks**, but give **targeted, time-limited** support to the vulnerable.

Risks that remain

- **Long-lasting high prices** raise costs and inflation and slow growth.
- **Blanket subsidies** can undo fiscal consolidation.
- **Capital outflows** in global risk-off phases pressure the **rupee**; heavy borrowing by rich countries and the **AI investment boom** may **pull capital away** from emerging markets.
- **CAD** keeps India dependent on **foreign financing**; **uneven State governance**; resistance to reforms such as **fertiliser pricing**; markets **overestimating** India's oil risk.

Way forward

- Speed up the **green transition** and **energy efficiency**.
- **Diversify trade**, sign **FTAs** and lift exports to shrink the CAD.
- **Fertiliser reform** linked to **soil health and balanced nutrients**; stronger **cooperative and competitive federalism**.
- **Clear communication** of how fiscal, monetary and supply-side steps fit together – turning **resilience into competitive advantage**.
- **Value addition:** India imports about **85–88%** of its crude; **Strategic Petroleum Reserves** of about **5.33 million tonnes** are at **Visakhapatnam, Mangaluru and Padur**.

PYQ – PRELIMS

- **Pattern:** Prelims asks about the **CAD, inflation targeting, forex reserves** and the **Strategic Petroleum Reserve**.

PYQ – MAINS

- **Pattern:** Mains asks about **energy security**, **external shocks** and the **fiscal–monetary policy mix**.

FQ – PRELIMS (Possible Question)

- India's Strategic Petroleum Reserves are located at: 1. Visakhapatnam 2. Mangaluru 3. Padur 4. Kochi. Select the correct answer: (a) 1, 2 and 3 only (b) 2 and 4 only (c) 1 and 4 only (d) 1, 2, 3 and 4 **(Ans: a)**

FQ – MAINS (Possible Question)

- "India should move from absorbing oil shocks to using them as a trigger for structural reform." Discuss with reference to macroeconomic policy and energy security. (250 words)

GS-III | Internal & Maritime Security – Defence Modernisation

11. Project 18A: Green Light for India's Next-Generation Destroyers

Syllabus: *GS-III: Security challenges and their management; defence technology and indigenisation.*

CONTEXT

The **Defence Acquisition Council (DAC)**, chaired by the **Defence Minister**, gave **in-principle approval** to the Navy's **Project 18A – Next-Generation Destroyers** worth about **₹80,000 crore**.



Photo: INS Visakhapatnam (left), lead ship of Project 15B, on exercise in the Indian Ocean – Project 18A ships will succeed this class.

Photo: U.S. Navy photo by Mass Communication Specialist 2nd Class Andrew Benvie, Wikimedia Commons, Public domain

KEYWORDS DEFINED

Keyword	Meaning
Destroyer	A **large, fast, heavily armed warship** that protects fleets against **air, surface and submarine** threats.
Vertical Launch System (VLS)	**Missile cells set into the deck** that fire missiles **straight up**, allowing many missiles in a small space.
AESA radar	**Active Electronically Scanned Array** – a radar that steers its beam **electronically**, tracking many targets at once.
Acceptance of Necessity (AoN)	The **DAC's first approval** that a defence purchase is **needed**, under the **Defence Acquisition Procedure**.
Integrated electric propulsion	Ships whose engines **generate electricity** that drives the **propellers and all systems**.

What Project 18A is

- The Navy's **Next-Generation Destroyer (NGD)** programme – **large, stealthy guided-missile destroyers**.
- Will succeed the **Visakhapatnam class (Project 15B)** as the Navy's main **surface strike** ships.
- **Designed by:** the Navy's in-house **Warship Design Bureau (WDB)**.
- **Aim:** multi-role ships for **land attack, fleet air defence and anti-submarine warfare (ASW)** across the **Indian Ocean Region**.

Key features

- **Size:** about **10,000–13,000 tonnes** – bigger than the Visakhapatnam class – with **stealth shaping** and **integrated electric propulsion**.
- **Indigenous content:** about **75%** – hull, electronics and weapons – under **Make in India / Aatmanirbhar Bharat**.
- **Firepower:** up to **144 VLS cells** for **BrahMos**, **long-range land-attack missiles** and **air-defence interceptors**.
- **Sensors:** indigenous **AESA radars** and an **integrated combat management system**.
- **ASW:** advanced **sonars**, the **SMART** system and **Varunastra** torpedoes.

Significance

- Strengthens India's **blue-water navy** against expanding **Chinese naval presence** and **submarine threats** in the IOR.
- Replaces **ageing destroyers**, sustaining **fleet strength** and **shipbuilding jobs**.

Value addition

- **Project 15B** ships: **INS Visakhapatnam, Mormugao, Imphal and Surat**, built by **Mazagon Dock (MDL), Mumbai**.
- **SMART** = **Supersonic Missile Assisted Release of Torpedo** (DRDO); **Varunastra** is a heavyweight torpedo from **NSTL, Visakhapatnam**.
- The **DAC** is the Defence Ministry's top body for **capital acquisition**; its approval is the **first step**, not a final contract.

PYQ – PRELIMS

- **Pattern:** Prelims asks about **naval projects, missiles and indigenous defence systems** (e.g., BrahMos, Project 75, INS Vikrant).

PYQ – MAINS

- **Pattern:** Mains asks about **indigenisation of defence** and India's **maritime security** in the Indian Ocean.

FQ – PRELIMS (Possible Question)

- Consider: 1. Project 18A destroyers will succeed the Visakhapatnam class. 2. SMART stands for Supersonic Missile Assisted Release of Torpedo. 3. Varunastra is a surface-to-air missile. Which are correct? **(Ans: 1 and 2 only)**

FQ – MAINS (Possible Question)

- "Indigenous warship building is central to India's ambition of being a net security provider in the Indian Ocean." Discuss with reference to Project 18A. (150 words)

GS-III | Agriculture – High-Value Crops

12. Saffron Blooms in Ladakh: A New Frontier for the 'Red Gold'

Syllabus: *GS-III: Agriculture – cropping patterns, high-value crops; agriculture in hill and cold-arid regions.*

CONTEXT

A **pilot saffron plot** on barren land revived under **Project Him Sarovar** in **Ladakh** flowered in just **30 days** – about **half the usual two months**.



Photo: workers picking purple saffron flowers at dawn on a saffron farm – only the red stigmas become the spice.
Photo: Safa.daneshvar, Wikimedia Commons, CC BY-SA 3.0

KEYWORDS DEFINED

Keyword	Meaning
Crocus sativus	The **saffron crocus**, a perennial of the **iris family (Iridaceae)**.
Stigma	The **thread-like female part of the flower**; saffron is the **dried red stigmas**.
Corm	An **underground swollen stem** that stores food and is used to **propagate** the plant.
Triploid	Having **three sets of chromosomes**; such plants are often **sterile** and cannot set seed.
Geophyte	A plant that survives harsh seasons as an **underground organ** (bulb, corm or tuber).

The crop

- Saffron comes from **Crocus sativus**, an **autumn-flowering geophyte**; each purple flower has **three crimson stigmas** – the world's **costliest spice**.
- **Where it grows: temperate and cold-arid** zones at about **1,500–2,800 m**.
- **Soil: well-drained, loose, neutral to slightly alkaline** sandy-loam or clay-loam – waterlogging causes **corm rot**.
- **Climate: warm, dry summers** for dormancy and **cool, sunny autumns** for flowering.

Special features

- **No seeds:** a **sterile triploid**, so it spreads only through **corms**, planted in late summer, which multiply each year.
- **Short bloom:** just **2–3 weeks** in **late October to mid-November**; flowers are **hand-picked at dawn**.
- **Labour-intensive:** about **1.5–1.75 lakh flowers** make **one kilogram** of dried saffron.
- **Chemistry: crocin** (colour), **picrocrocin** (bitter taste) and **safranal** (aroma).
- **Hardy:** needs **little water** and survives **snow and sub-zero** winters while dormant.

Why the Ladakh pilot matters

- Turns **cold, barren land** into **high-value farming** for marginal farmers.
- Spreads saffron **beyond Kashmir** – to **Ladakh, Himachal Pradesh and Sikkim** – reducing dependence on one region.
- Shows how **Him Sarovar water-storage** work can support new crops.

Value addition

- **Kashmir saffron** has a **GI tag (2020)**; **Pampore** is called India's **'saffron bowl'**.
- The **National Saffron Mission (2010)** aimed to revive Kashmir's saffron farms.
- **Iran** produces most of the **world's saffron**.

PYQ – PRELIMS

- **Pattern:** Prelims asks about **GI-tagged products**, **crop climatic needs** and **plant propagation** methods.

PYQ – MAINS

- **Pattern:** Mains asks about **diversifying agriculture** in hill and cold-desert regions and **high-value crops** for farm incomes.

FQ – PRELIMS (Possible Question)

- Consider: 1. Saffron is obtained from the dried stigmas of Crocus sativus. 2. Saffron is mainly propagated through seeds. 3. Safranal is responsible for the aroma of saffron. Which are correct? **(Ans: 1 and 3 only)**

FQ – MAINS (Possible Question)

- "High-value crops can transform cold-arid regions." Discuss with reference to the saffron pilot in Ladakh. (150 words)

GS-III | Agriculture – Cereals, Feed and Biofuels

13. 1st International Maize Conference: From 4F to 5F, Farmers First

Syllabus: *GS-III: Major crops and cropping patterns; agricultural research; biofuels and food security.*

CONTEXT

The **1st International Maize Conference (IMC 2026)** began in **Hyderabad** on the theme **'Maize Revolution for Food, Feed, Fodder, Fuel and Sustainability in the Changing World' (M4FS)**.



Photo: a maize crop in rural India – a grain used for food, poultry feed, fodder and ethanol.
Photo: Phil Bus, Wikimedia Commons, CC BY-SA 2.0

KEYWORDS DEFINED

Keyword	Meaning
4F / 5F	Maize's four uses – **Food, Feed, Fodder, Fuel** – now with a fifth F, **Farmers**.
Yield gap	The difference between **what farmers actually harvest** and **what is possible** with the best practices.
Aflatoxin	A **toxic, cancer-causing chemical** made by moulds on badly stored grain.
AICRP	**All India Coordinated Research Project** – ICAR's network for multi-location crop research.
Biofortification	Breeding crops with **more vitamins or minerals** in the grain.

About IMC 2026

- A global meeting of **scientists, policymakers, universities and agribusiness**, held in **Hyderabad, Telangana**.
- **Aim:** close **yield gaps**, breed **climate-resilient and biofortified hybrids**, and meet rising demand for **ethanol, poultry feed and fodder**.

Key outcomes

- **4F → 5F: Farmers** added to Food, Feed, Fodder and Fuel – focus on **profit and lower input costs**.
- **Target:** raise production from about **55 to 75 million tonnes by 2036** through better **hybrids and crop management**.
- **Zoning without displacement:** grow more maize in **kharif, rabi and spring** seasons **without replacing pulses and oilseeds**.

- **Seeds and bio-refineries:** strengthen the **AICRP on Maize**, more **public hybrid seed**, **aflatoxin-free grain**, and support for **20% ethanol blending**.

India and maize

- **Area:** India is **4th in the world** (after the **USA, China and Brazil**).
- **Output:** among the **top five or six producers**, about **3–4%** of world output.
- **Leading States: Karnataka, Madhya Pradesh and Maharashtra**, followed by **Bihar, Rajasthan, Uttar Pradesh and Telangana**.

Value addition

- Maize is called the **'queen of cereals'** for its high **yield potential**; it is a **C4 plant** that uses water and sunlight efficiently.
- The **ICAR-Indian Institute of Maize Research** is at **Ludhiana**.
- Grain-based ethanol has made maize a key **feedstock** for **E20** blending – raising concerns about **feed prices for poultry**.

PYQ – PRELIMS

- **Pattern:** Prelims asks about **crops and their uses**, **biofuels**, and **agricultural research bodies**.

PYQ – MAINS

- **Pattern:** Mains asks about **crop diversification**, **food-versus-fuel** trade-offs and **farm incomes**.

FQ – PRELIMS (Possible Question)

- Consider: 1. India ranks fourth in the world by area under maize. 2. Maize is a C3 plant. 3. Aflatoxin is a toxin produced by moulds on stored grain. Which are correct? **(Ans: 1 and 3 only)**

FQ – MAINS (Possible Question)

- "Maize can serve food, feed, fodder and fuel – but only if farmers profit." Discuss the challenges of expanding maize output in India. (150 words)

GS-III | Science & Technology – Space and Astronomy

14. Ultraluminous X-ray Sources: Black Holes Shining Past the Limit

Syllabus: *GS-III: Science and technology – awareness in the field of space.*

CONTEXT

Astronomers who followed the source **ULX-8 in the galaxy NGC 4490** for **24 years** saw changes in its **X-ray behaviour** and found evidence that a **stellar-mass black hole** feeding **at or near the Eddington limit** powers it.



Photo: the interacting galaxies NGC 4490 and NGC 4485 (the 'Cocoon Galaxy' pair) – home of the source ULX-8.
Photo: KPNO/NOIRLab/NSF/AURA/Michael Gariepy/Adam Block, Wikimedia Commons, CC BY 4.0

KEYWORDS DEFINED

Keyword	Meaning
Ultraluminous X-ray Source (ULX)	A **very bright X-ray source** outside a galaxy's centre that gives off more X-rays than **a million Suns** give off in all forms of light.
Eddington limit	The brightness at which the **outward push of radiation** equals the **inward pull of gravity**.
Accretion	The process of **matter falling onto a compact object**, heating up and glowing.
Stellar-mass black hole	A black hole formed from a **collapsed star**, typically a few to tens of **solar masses**.
Neutron star	The **extremely dense core** left after a massive star explodes as a supernova.

What ULXs are

- **Extremely bright X-ray sources** in **nearby galaxies**, emitting more X-rays than **a million Suns** emit across all wavelengths.
- Powered by **neutron stars or black holes** that **pull in matter** from a **companion star**.
- **Rare:** most galaxies, including the **Milky Way**, have **none**; those that do usually have **one**.
- The ring galaxy **NGC 922** hosts **seven** ULXs, found with NASA's **Chandra X-ray Observatory**.

Why they puzzle astronomers

- They look **brighter than the classical Eddington limit** should allow.
- **Above that limit**, radiation should **blow away the gas** feeding the object and stop the glow.
- Possible answers: **beamed radiation**, **very fast accretion** onto neutron stars, or rare **intermediate-mass black holes**.

The ULX-8 finding

- Watching **NGC 4490 ULX-8 for 24 years** revealed changes in its **X-ray output**.
- The data fit a **stellar-mass black hole** feeding **at or near the Eddington limit** – helping explain how ULXs shine so brightly.

Value addition

- **Chandra (NASA, 1999)** is a leading X-ray telescope; X-rays are **absorbed by Earth's atmosphere**, so they must be studied **from space**.
- **India: AstroSat (2015)** – India's first **multi-wavelength space observatory** – and **XPoSat (2024)**, its first **X-ray polarimetry** mission.

PYQ – PRELIMS

- **Pattern:** Prelims asks about **black holes, neutron stars, gravitational waves** and **space observatories**.

PYQ – MAINS

- **Pattern:** Mains asks about India's **space science missions** and their value.

FQ – PRELIMS (Possible Question)

- Consider: 1. Most galaxies, including the Milky Way, host several ULXs. 2. The Eddington limit is reached when radiation pressure balances gravity. 3. XPoSat is India's X-ray polarimetry mission. Which are correct? **(Ans: 2 and 3 only)**

FQ – MAINS (Possible Question)

- "Space-based astronomy opens windows that ground telescopes cannot." Discuss with reference to X-ray astronomy and India's missions. (150 words)

GS-III | Environment – Air and Noise Pollution

15. Green Crackers: Celebrating Diwali With Less Smoke and Noise

Syllabus: *GS-III: Environment – pollution and its control; S&T applications. GS-II: Government policies (enforcement).*

CONTEXT

The **Chandigarh administration** has allowed **only certified green crackers** for **Diwali**, part of wider moves to cut **firecracker pollution**.



Illustration: a light festive burst and thin smoke, with heavy black smoke crossed out – the idea behind green crackers.

KEYWORDS DEFINED

Keyword	Meaning
Green crackers	**Low-emission firecrackers** that release **less smoke, dust and noise** than conventional ones.
Particulate matter (PM)	**Tiny solid or liquid particles** in the air (PM2.5, PM10) that damage the **lungs and heart**.
CSIR-NEERI	**National Environmental Engineering Research Institute**, Nagpur – the CSIR lab that developed green crackers.
PESO	**Petroleum and Explosives Safety Organisation** – the national regulator for explosives, including fireworks.
Decibel (dB)	The unit used to measure **loudness of sound**.

What green crackers are

- Firecrackers designed to **cut air and noise pollution** while keeping celebrations **safer**.
- **Developed by:** scientists at **CSIR-NEERI**.
- **Aim:** reduce **particulate matter and gases** released when crackers burst.

Composition (as cited)

- About **29% aluminium, 28% potassium nitrate, 38%** of a **CSIR-NEERI proprietary material** and **5% sulphur**.
- **Banned or reduced chemicals:** no **barium nitrate, lithium, mercury, arsenic or cadmium**; less of some raw materials.

Key features

- About **30% less particulate matter**, and at least **10% fewer gaseous emissions**.
- **Noise limit:** below **125 decibels at 4 metres**.
- **Identification:** certified boxes carry **CSIR-NEERI and PESO logos** and an **encrypted QR code**.

Value addition

- Types developed by NEERI include **SWAS** (safe water releaser), **STAR** (safe thermite cracker) and **SAFAL** (safe minimal aluminium).
- In the **Arjun Gopal** case, the **Supreme Court** allowed only **green crackers** and restricted **barium-based** ones.
- **PESO** works under the **Department for Promotion of Industry and Internal Trade (DPIIT)**.
- **Limits:** 'green' crackers still pollute – **less is not none**; enforcement against **fake QR codes** is key.

PYQ – PRELIMS

- **Pattern:** Prelims asks about **air pollutants, CSIR labs** and **regulators** of hazardous substances.

PYQ – MAINS

- **Pattern:** Mains asks about **urban air pollution**, **seasonal smog** and balancing **culture with public health**.

FQ – PRELIMS (Possible Question)

- Consider: 1. Green crackers were developed by CSIR-NEERI. 2. Certified green crackers carry CSIR-NEERI and PESO logos with a QR code. 3. Green crackers produce no particulate matter at all. Which are correct? **(Ans: 1 and 2 only)**

FQ – MAINS (Possible Question)

- "Festivals and clean air need not be opposites." Discuss the role of green crackers and enforcement in tackling festive pollution. (150 words)

GS-III | Indian Economy – Capital Markets

16. IPO Grey Market: The Unofficial Price Before Listing

Syllabus: *GS-III: Indian economy – capital markets and their regulation; investor protection.*

> **CONTEXT**
> India's **IPO boom** has fuelled **unofficial 'grey market' trading**, where shares change hands **before listing** and the **grey market premium (GMP)** is used to guess **listing-day gains**.



Illustration: a share certificate passed in a shadowy corner outside a bright stock exchange – unofficial trading with question marks.

KEYWORDS DEFINED

Keyword	Meaning
IPO	**Initial Public Offering** – a company's **first sale of shares** to the public.
Issue price	The **price at which IPO shares are offered** to investors.
Grey Market Premium (GMP)	The **extra amount** over the issue price at which IPO shares are **informally traded** before listing.
Listing gain	The **profit** when a share opens **above its issue price** on listing day.
Kostak	Grey-market slang for a **price paid for an IPO application** itself, whether or not shares are allotted.

What the grey market is

- An **unofficial market** where IPO shares are **traded before they list** on a recognised stock exchange.
- Works through **informal deals** between buyers and sellers, usually **without paperwork**; often **no shares actually change hands** – people **bet on the listing price**.
- **Example:** issue price **₹1,000** and GMP **₹300** → informal value **₹1,300**.
- Said to be run by players in **Ahmedabad, Jaipur and Rajkot** (as cited).

Why it matters

- **Not regulated by SEBI** and outside **stock exchanges**.
- **GMP is not official** and **not a guarantee** of the listing price – yet it **sways investor mood**.

Risks

- **No assured profit:** the listing price may be **higher or lower** than the GMP suggests.
- **Bad decisions:** investors may skip studying a company's **finances, valuation and prospects**.
- **No transparency:** prices, demand and deals **cannot be verified**.
- **Manipulation:** premiums can be **pumped up** to create hype.
- **Small investors** may put in **savings or borrowed money** chasing listing gains.

Value addition

- SEBI cut the **IPO listing timeline to T+3** (three working days after the issue closes) in **2023**.
- SEBI has **warned investors** against relying on GMP and has discussed a regulated **'when-listed' pre-listing** window (as reported).
- **Safer path:** read the **prospectus (RHP)**, check **valuation and promoters**, and use **ASBA** through a bank.

PYQ – PRELIMS

- **Pattern:** Prelims asks about **capital-market terms** – IPO, FPO, ASBA, SEBI's role.

PYQ – MAINS

- **Pattern:** Mains asks about **investor protection** and **financialisation of household savings**.

FQ – PRELIMS (Possible Question)

- With reference to the grey market for IPOs, consider: 1. It is regulated by SEBI. 2. The grey market premium guarantees the listing price. 3. Deals are usually informal agreements outside stock exchanges. Which are correct? **(Ans: 3 only)**

FQ – MAINS (Possible Question)

- "The IPO boom has drawn millions of new investors – and new risks." Discuss the role of the grey market and the need for investor protection. (150 words)

GENERAL STUDIES PAPER IV

[GS-IV] Ethics, Integrity & Aptitude

GS-IV | Ethics, Integrity & Aptitude – Synthesis

17. Ethics Synthesis: Consent, Care and Accountability in a Digital Age

Syllabus: *GS-IV: Ethics and human interface; probity in governance; information sharing and transparency; environmental ethics; case studies.*

CONTEXT

Today's topics share one thread: **power must come with responsibility**. **AI firms using our data**, **children online**, **rural workers seeking redress**, **festive pollution**, **speculative markets** and a **melting cryosphere** all test **consent, care and accountability**.



Illustration: a balance weighing care, safety and nature against money and haste.

KEYWORDS DEFINED

Keyword	Meaning
Informed consent	Agreement given **freely and with full knowledge** of what will happen with one's data or person.
Duty of care	A duty to **avoid harm** to those who **depend on you** – especially children and the vulnerable.
Accountability	Being **answerable** for decisions and their results.
Intergenerational equity	Fairness between **present and future generations** in using resources.
Harm principle	**J.S. Mill's** idea that freedom may be limited **only to prevent harm to others**.

The ethical thread

- **Privacy and consent:** AI training on **personal data** without people knowing treats them as **means, not ends**.
- **Duty of care:** teachers trained to protect **children online** show **care beyond the classroom**.
- **Accountability: time-bound grievance redressal** makes officials answer to **rural workers**.

- **Freedom vs harm: green crackers** balance **celebration** with **others' right to clean air**.
- **Prudence and honesty:** the **grey market** feeds on **greed and hype**; regulators and investors need **restraint and truthfulness**.
- **Stewardship:** the **cryosphere report** and **Shey-Phoksundo** ask us to hold nature **in trust for future generations**.

Values at stake

- **Respect for persons, compassion, accountability, transparency, prudence, justice, stewardship, scientific temper.**

Case study 1: The scraped dataset

- **Situation:** You are a **data scientist** at an Indian AI start-up. Your manager wants to train a model on **millions of scraped social-media posts**, including **personal photos**, before a funding deadline.
- **Issues: privacy and consent vs speed and profit**; legal grey zone; trust of users.
- **Options:** (a) go ahead – the data is 'public'; (b) quietly quit; (c) **raise the issue**, propose **filtering personal data**, **anonymisation and opt-outs**, and record your objection.
- **Preferred course: (c)** – legality is the floor, **ethics is the standard**; **privacy by design** protects both users and the firm.

Case study 2: The message in the class group

- **Situation:** As a **school principal**, you learn that a **Class 8 student** is being messaged by an unknown adult on a gaming app, asking for photos. The parents fear 'bad name' and want it hushed up.
- **Issues: child safety vs family reputation**; legal duty to report; confidentiality.
- **Options:** (a) keep it quiet as the parents ask; (b) punish the student for using the app; (c) **protect the child**, **report under POCSO** and on **1930/cybercrime.gov.in**, involve a **counsellor**, and run **awareness sessions** without naming the child.
- **Preferred course: (c)** – the **child's welfare is paramount**; silence protects the offender.

Case study 3: The delayed wage complaints

- **Situation:** As a **Block Grievance Redressal Officer**, you find **200 wage complaints** pending. A local leader asks you to **close them as 'resolved'** before an inspection.
- **Issues: integrity vs pressure**; workers' right to wages; the new **15-day rule**.
- **Options:** (a) close them on paper; (b) wait for the inspection to pass; (c) **prioritise by age**, set up **camp-mode hearings**, **escalate** what you cannot solve, and **record the pressure**.
- **Preferred course: (c)** – **accountability to citizens**, not to influence.

Thinkers to quote (paraphrased)

- **Immanuel Kant:** treat people as **ends in themselves**, never merely as **means** – including as data.
- **J.S. Mill:** liberty ends where **harm to others** begins – the logic behind cracker limits.
- **Mahatma Gandhi:** 'The earth has **enough for everyone's need**, but not for everyone's greed.'
- **John Rawls:** design rules from behind a **'veil of ignorance'** – as if you could be the **most vulnerable** person.
- **Aristotle: practical wisdom** – the **right action, at the right time, in the right way**.

PYQ – PRELIMS

- **Not applicable** – GS-IV is Mains-only.

PYQ – MAINS

- **Pattern:** GS-IV regularly asks about **privacy, data ethics**, **protection of children**, **accountability** and **case studies** on administrative pressure.

FQ – PRELIMS (Possible Question)

- **Not applicable** – GS-IV is Mains-only.

FQ – MAINS (Possible Question)

- "Consent is the foundation of ethical data use." Discuss with reference to AI and the right to privacy. (150 words)
- Case study: As a principal, how would you handle an online grooming attempt on a student when the family wants silence? (250 words)